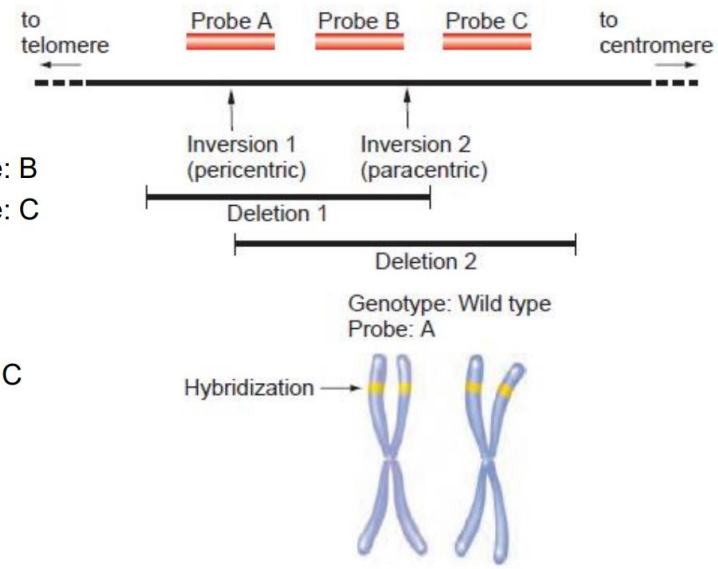
Problems

1. Human chromosome 1 is a large, metacentric chromosome. A map of a cloned region from near the telomere of chromosome 1 is shown below. Three probe DNAs (A, B, and C) from this region were used for *in situ* hybridization to human mitotic metaphase chromosome squashes made with cells obtained from individuals with various genotypes. The breakpoints of chromosomal rearrangements in this region are indicated on the map. The black bars for deletions (*Del*) 1 and 2 represent DNA that is deleted. The breakpoints of inversions (*Inv*) 1 and 2 not shown in the figure are near but not at the centromere. For each of the following genotypes, draw chromosome 1 as it would appear after *in situ* hybridization. An example is shown in the following figure for hybridization of probe A to the two copies of chromosome 1 in wild type(/).

a. genotype: *Del1/Del2;* probe: B
b. genotype: *Del1/Del2;* probe: C
c. genotype: *Del1/ ;* probe: A
d. genotype: *Inv1 / ;* probe: A
e. genotype: *Inv2 / ;* probe: B
f. genotype: *Inv2/Inv2;* probe: C

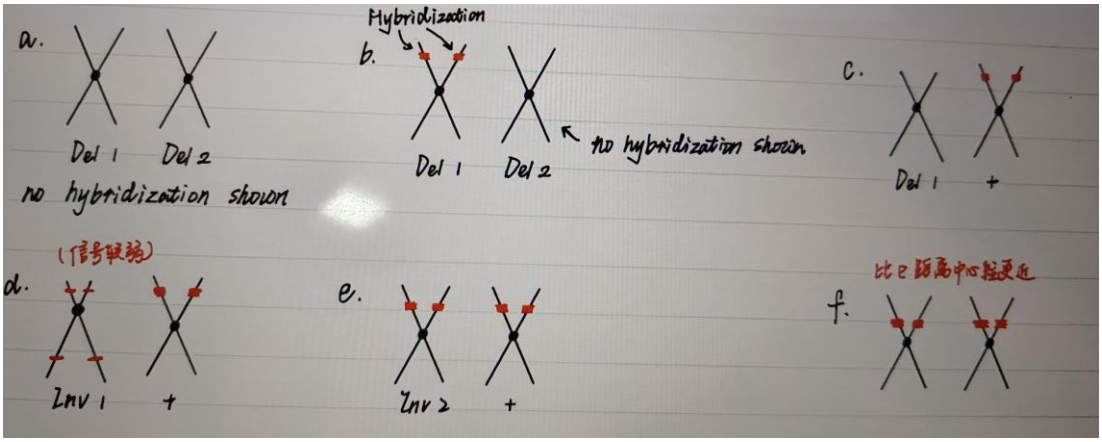


Problems

2. Design a simple method to detect Prader-Willi Syndrome, such as SNP detection or copy number detection or other smart methods you have. Please write the relevant sequence (including primers, SNPs) for your detection. Some methods are listed in

An Overview of Mutation Detection Methods in Genetic Disorders. Iran J Pediatr. 2013 Aug; 23(4): 375–388.

1.解答:



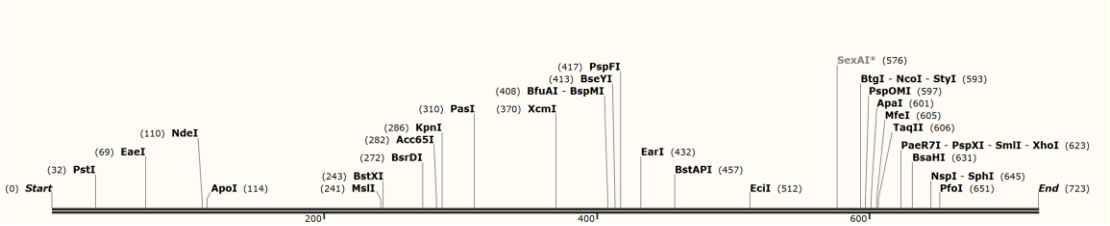
2.解答:

Prader-Willi syndrome is due to absence of paternally expressed imprinted genes at 15q11.2-q13 through paternal deletion of this region (65–75% of individuals), maternal uniparental disomy 15 (20–30%), or an imprinting defect (1–3%). Parent-specific DNA methylation analysis will detect >99% of individuals.

染色体 15q11.2-13 位置的小核核糖核蛋白多肽 N 基因(small nuclear ribonucleoprotein polypeptide N，SNRPN）是该综合征的一个重要基因，在正常人母源性 SNRPN 的 CpG 岛高度甲基化，而父源 SNRPN 的 CpG 岛没有甲基化，因此仅来自父亲的基因有表达，而母亲来源的基因失活。经过亚硫酸盐修饰，父源 DNA 上的非甲基化胞嘧啶转变为尿嘧啶，而母源甲基化胞嘧啶不变，再设计特异性引物进行 PCR 就可将等位基因分离。

引物设计:

NCBI 查阅到 SNRBP 的序列



母源选取其中约 200bp 进行 PCR

F: 5'-GCTGCTGCTGTTGCTGCGA-3'

R: 3'-CGGTTATCCGTAGGCGGAG-5'

父源选取其中约 150bp 进行 PCR

F: 5'- GTTCAGAAAGATCAAGCCA-3'

R: 3'- CCATGGTGAACGACCTCG-5'

正常人电泳结果为母源条带和父源条带；确诊患者仅显示母源条带。

Reference：

1. Cassidy SB, Schwartz S, Miller JL, Driscoll DJ. Prader-Willi syndrome. Genet Med. 2012 Jan;14(1):10-26. doi: 10.1038/gim.0b013e31822bead0. Epub 2011 Sep 26. PMID: 22237428.
2. 王薇, et al. (2008). "甲基化特异性 PCR 方法诊断 Prader-Willi 综合征." 中国当代儿科杂志 (04): 485-488.
3. Prader-Willi 综合征的快速诊断，张惠文、叶军，2018. https://www.sohu.com/a/274969770_100281680